



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

Go, change the world

Federated Heterogeneous Graph Neural Networks for TBML Detection

A Major Project Report (CS481P)

Submitted by,

Nikhil Vasu

1RV22CS128

Rohith Biradar

1RV22CS164

Suraj Chanaveeragoudra

1RV22CS211

Under the guidance of

Dr. Nagaraj G. S

Prof. Sanjana Ravindra Otihal

Professor

Assistant Professor

Dept. of CSE

Dept. of CSE

RV College of Engineering

RV College of Engineering

In partial fulfillment of the requirements for the degree of

Bachelor of Engineering in

Computer Science and Engineering

2025-26

RV College of Engineering[®], Bengaluru
(*Autonomous institution affiliated to VTU, Belagavi*)
Department of Computer Science and Engineering



CERTIFICATE

Certified that the major project (CS481P) work titled ***Federated Heterogeneous Graph Neural Networks for TBML Detection*** is carried out by **Nikhil Vasu (1RV22CS128)**, **Rohith Biradar (1RV22CS164)** and **Suraj Chanaveeragoudra (1RV22CS211)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major project report deposited in the departmental library. The major project report has been approved as it satisfies the academic requirements in respect of major project work prescribed by the institution for the said degree.

Guide

Head of the Department

Principal

Dr. Nagaraj G. S

Dr. Shanta Rangaswamy

Dr. K. N. Subramanya

External Viva

Name of Examiners

Signature with Date

1.

2.

DECLARATION

We, **Nikhil Vasu**, **Rohith Biradar** and **Suraj Chanaveeragoudra** students of eighth semester B.E., Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**Federated Heterogeneous Graph Neural Networks for TBML Detection**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Computer Science and Engineering** during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Nikhil Vasu(1RV22CS128)
2. Rohith Biradar(1RV22CS164)
3. Suraj Chanaveeragoudra(1RV22CS211)

ACKNOWLEDGEMENTS

We are indebted to our guide, **Dr. Nagaraj G. S**, Professor, Department of CSE, RVCE for the wholehearted support, suggestions and invaluable advice throughout our project work and also helped in the preparation of this thesis.

We also express our gratitude to our panel members **Dr. Nagaraj G. S**, Professor, Department of CSE, RVCE and **Prof. Sanjana Ravindra Otihal**, Assistant Professor, Department of CSE, RVCE for the valuable comments and suggestions during the phase evaluations.

Our sincere thanks to the project coordinators **Dr. Chethana R Murthy**, Professor, Department of CSE and **Prof. Deepika Dash**, Assistant Professor, for their timely instructions and support in coordinating the project.

Our gratitude to **Prof. Narashimaraja P**, Department of ECE, RVCE for the organized latex template which made report writing easy and interesting.

Our sincere thanks to **Dr. Shanta Rangaswamy**, Professor and Head, Department of Computer Science and Engineering, RVCE for the support and encouragement.

We express sincere gratitude to our beloved Professor and Vice Principal, **Dr. Geetha K S**, RVCE and Principal, **Dr. K. N. Subramanya**, RVCE for the appreciation towards this project work.

We thank all the teaching staff and technical staff of Computer Science and Engineering department, RVCE for their help.

Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

Trade-Based Money Laundering (TBML) has become a major challenge in modern financial systems due to the increasing complexity and scale of global trade transactions. Conventional anti-money laundering systems primarily rely on rule-based monitoring and traditional machine learning techniques, which are often inadequate for identifying hidden transactional dependencies and evolving laundering strategies. These approaches suffer from limitations such as high false positive rates, poor relational learning capability, and reduced scalability in large financial networks. To address these issues, this report presents a Federated Heterogeneous Graph Neural Network (HGNN) framework for real-time TBML detection capable of modeling complex trade relationships while preserving institutional data privacy.

The primary objective of this work is to develop a scalable and privacy-preserving TBML detection architecture using graph-based deep learning methodologies. The proposed framework represents financial ecosystems as heterogeneous graphs consisting of entities such as accounts, importers, exporters, transactions, and trade documents. Graph Attention Network (GAT)-based convolutional layers are employed to learn structural and semantic relationships between interconnected entities for effective anomaly detection. Federated learning mechanisms are incorporated to facilitate collaborative model training across distributed financial institutions without centralized data sharing. The computational framework includes graph construction, feature extraction, node embedding generation, suspicious activity classification, and real-time risk score prediction.

The proposed system is implemented using Python, PyTorch Geometric, FastAPI, Memgraph, PostgreSQL, and React.js. Experimental evaluation is carried out on simulated TBML datasets containing both legitimate and suspicious trade transactions. Performance analysis is conducted using standard classification metrics including accuracy, precision, recall, and F1-score. The experimental results indicate that the proposed framework effectively captures hidden laundering patterns and improves fraud detection capability when compared with conventional machine learning approaches. The architecture further supports scalable deployment, reduced false positive rates, and efficient real-time monitoring of high-volume trade transactions.

CONTENTS

LIST OF FIGURES

LIST OF TABLES

Chapter 1

Introduction to Trade-Based Money Laundering

CHAPTER 1

INTRODUCTION TO TRADE-BASED MONEY LAUNDERING

The rapid growth of international trade and digital financial infrastructure has significantly increased the complexity of monitoring global financial transactions. Alongside economic globalization, financial institutions are facing increasingly sophisticated forms of illicit financial activities that exploit weaknesses in conventional Anti-Money Laundering (AML) systems. Among these, Trade-Based Money Laundering (TBML) has emerged as one of the most challenging financial crimes due to its ability to conceal illegal fund transfers within legitimate trade operations. The distributed and interconnected nature of modern trade ecosystems necessitates intelligent, scalable, and privacy-preserving analytical frameworks capable of identifying hidden transactional relationships in real time.

“For financial institutions, TBML is uniquely challenging because illicit funds are embedded within legitimate trade transactions, leaving traditional AML controls blind to the risks.”

– Dr. Sebastian Hetzler, IMTF Co-CEO [1]

As highlighted in the above statement, Trade-Based Money Laundering exploits the complexity of international commerce by embedding illicit financial flows within apparently legitimate trade transactions. This significantly limits the effectiveness of conventional compliance systems that primarily rely on isolated transaction monitoring and static rule-based verification mechanisms.

1.1 Overview of Trade-Based Money Laundering

Trade-Based Money Laundering (TBML) is defined as the process of disguising the proceeds of crime and transferring value through legitimate international trade transactions in an attempt to legitimize illicit financial activities [2]. Unlike conventional money laundering mechanisms that rely on direct movement of funds, TBML conceals illegal financial flows within normal commercial trade operations such as import-export activities, invoice settlements, and cross-border transactions. Due to the involvement of multiple intermediaries, financial institutions, customs agencies, and international jurisdictions,

identifying suspicious trade behavior becomes significantly difficult for conventional Anti-Money Laundering systems.

1.1.1 Global Financial Crime Scenario

The increasing volume of international trade and cross-border financial transactions has considerably complicated the process of monitoring illicit financial activities. According to the United Nations Office on Drugs and Crime (UNODC), money laundering activities account for nearly 2% to 5% of the global Gross Domestic Product (GDP), corresponding to approximately \$800 billion to \$2 trillion annually [3]. Despite the deployment of large-scale compliance frameworks and transaction monitoring systems, Europol reports that global authorities successfully intercept less than 1% of illicit financial flows [4]. These statistics indicate the growing sophistication of modern laundering networks and the limitations of existing monitoring infrastructures.

The large-scale movement of goods, services, and financial assets across international markets creates an environment where illicit transactions can be concealed within legitimate commercial activities. As financial institutions continue to strengthen Know-Your-Customer (KYC) regulations and banking surveillance systems, criminal organizations increasingly exploit trade ecosystems to bypass conventional financial monitoring mechanisms.

Table 1.1: Key Statistics Related to Global Trade-Based Financial Crime

Financial Crime Metric	Estimated Value
Global GDP associated with money laundering annually	2% – 5%
Estimated annual TBML economic volume	\$1.6T – \$2.0T
Illicit flows linked to trade misinvoicing in developing economies	87%
Global interception rate of illicit financial flows	<1%
Reduction in false positive alerts using AI-driven monitoring systems	Up to 40%

1.1.2 Trade-Based Money Laundering

Trade-Based Money Laundering enables criminal organizations to transfer illicit value through apparently legitimate trade transactions while avoiding direct movement of cash. Common TBML techniques include over-invoicing, under-invoicing, multiple invoicing,

false description of goods, phantom shipments, and the use of shell entities to manipulate commercial trade records. Since these activities are embedded within genuine trade operations involving multiple organizations and jurisdictions, detecting suspicious transactional behavior becomes highly complex for financial institutions and regulatory agencies.

Reports published by Global Financial Integrity indicate that trade misinvoicing and related TBML activities account for nearly 87% of illicit financial flows originating from developing economies [5]. Similarly, industry analyses estimate that approximately \$1.6 trillion to \$2 trillion in illicit value is concealed annually within trade invoices and commercial documentation [1]. These findings demonstrate the growing dependence of organized criminal networks on international trade systems for laundering illicit financial assets.

1.1.3 Problem Area

Conventional Anti-Money Laundering systems primarily rely on rule-based monitoring frameworks, predefined transaction thresholds, and manual auditing procedures for identifying suspicious financial activities. Although such approaches are effective in detecting isolated anomalies, they often fail to identify hidden relationships among entities involved in coordinated laundering operations. Modern TBML networks operate through interconnected ecosystems involving importers, exporters, shell companies, offshore entities, logistics providers, and financial intermediaries distributed across multiple jurisdictions, making detection significantly more challenging.

Traditional machine learning approaches further suffer from limitations associated with isolated transaction analysis and lack of relational understanding. Since financial transactions are processed independently, existing systems are unable to effectively capture structural interactions and behavioral dependencies hidden within large-scale trade networks. As a result, conventional AML systems frequently generate high false positive rates, increased compliance overhead, and delayed investigation processes.

Another major challenge involves institutional privacy and regulatory compliance constraints. Financial organizations are often unable to share sensitive transactional data across institutions due to strict legal and regulatory restrictions. Consequently, centralized fraud detection architectures become impractical in real-world banking environments. These limitations highlight the need for intelligent, scalable, and privacy-preserving analytical frameworks capable of effectively identifying suspicious transactional behavior

across distributed financial ecosystems.

1.2 Literature Review

The increasing complexity of global financial systems and the rapid growth of international trade have motivated extensive research in Anti-Money Laundering (AML) and Trade-Based Money Laundering (TBML) detection systems. Conventional AML frameworks primarily relied on rule-based transaction monitoring and manual auditing procedures to identify suspicious financial activities. However, the emergence of interconnected financial ecosystems, cross-border trade mechanisms, and sophisticated laundering strategies has significantly reduced the effectiveness of traditional approaches. Consequently, recent research has increasingly focused on integrating machine learning, graph analytics, and distributed intelligence techniques into modern AML systems.

Jiani Fan, Lwin Khin Shar, Ruichen Zhang, Ziyao Liu, Wenzhuo Yang, Dusit Niyato, and Kwok-Yan Lam [6] presented a comprehensive survey on deep learning-based Anti-Money Laundering frameworks for mobile financial systems. The study explored the application of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Graph Neural Networks (GNNs) for transaction anomaly detection and proposed a Context-Risk-Predict framework for improving AML intelligence. The authors demonstrated that deep learning architectures significantly improve predictive performance compared to conventional rule-based systems. However, the framework primarily focused on centralized transaction processing environments and did not address institutional privacy constraints or distributed financial data challenges.

Morten Hjelholt Jensen and Alexandros Iosifidis [7] investigated the use of statistical analysis and supervised machine learning algorithms for financial anomaly detection. Their study evaluated models such as Logistic Regression, Random Forests, and Gradient Boosting techniques for suspicious transaction classification. Experimental results showed improved detection capability compared to static rule-based approaches. Nevertheless, the proposed methods analyzed transactions independently and lacked the ability to capture relational dependencies among interconnected financial entities involved in coordinated laundering operations.

Jinlong Song, Zhen Guo, Yutian Lei, and Minghao Zhao [8] proposed a dimension expansion framework for learning hidden money laundering activities within transaction networks. The framework enhanced transaction representations by integrating neighbor-

ing account information and contextual interaction features. The study demonstrated that relational feature aggregation improves the detection of concealed laundering patterns across financial networks. However, the model depended heavily on manually engineered relational features and lacked scalability for distributed cross-institutional financial environments.

In another study, Jinlong Song, Haoran Chen, Peng Xu, and Minghao Zhao [9] introduced a Social Attention Graph Neural Network framework for detecting illicit accounts in blockchain transaction systems. The proposed architecture transformed transaction ecosystems into graph structures and utilized attention-based graph learning mechanisms for anomaly classification. The study demonstrated that Graph Neural Networks outperform conventional machine learning models in identifying suspicious transactional behavior. However, the framework focused primarily on cryptocurrency transaction ecosystems and did not address invoice-level trade manipulations or heterogeneous commercial trade relationships associated with TBML activities.

Xiang Li, Zhen Wang, Yong Zhou, and Lei Zhang [10] proposed a Multi-View Graph-Based Hierarchical Representation Learning framework for organized money laundering group detection. Their work utilized multiple graph perspectives and hierarchical embedding mechanisms to identify coordinated laundering communities within financial transaction networks. Experimental findings demonstrated improved graph representation capability for identifying hidden laundering structures. However, the framework relied on predefined graph meta-path relationships and centralized graph repositories, limiting its applicability in privacy-sensitive financial environments.

Eren Gezici and Emre Sefer [11] proposed AMLPD, an unsupervised Anti-Money Laundering detection framework utilizing Personalized PageRank and graph diffusion techniques for deep vertex representation learning. The framework effectively captured hidden graph structures without requiring labeled datasets and demonstrated strong anomaly detection capability in sparse financial networks. However, the absence of supervised optimization mechanisms resulted in increased false positive classifications within highly imbalanced financial transaction datasets.

Mohammad Karim, Tariq Hassan, and Sameer Ali [12] investigated scalable semi-supervised graph learning architectures for large-scale financial transaction monitoring. Their study evaluated models such as SkipGCN and EvolveGCN for node classification

within AML transaction graphs. Experimental results demonstrated efficient scalability and improved anomaly detection performance in large graph repositories. Nevertheless, the framework assumed centralized access to financial transaction graphs and did not address institutional privacy and regulatory constraints associated with collaborative fraud detection systems.

Jongmin Koo, Hyun Park, and Seungwoo Kim [13] proposed an autoencoder-based suspicious transaction detection framework using Risk-Based Approaches (RBA) for banking systems. The proposed model reduced anomaly detection complexity by learning latent transactional representations from financial datasets. Although the framework improved internal banking surveillance efficiency, it focused primarily on point anomalies associated with individual accounts and failed to model interconnected laundering relationships distributed across complex trade ecosystems.

The literature survey indicates a clear transition from conventional rule-based AML systems toward machine learning, graph neural networks, and intelligent graph analytics for financial crime detection. Existing studies demonstrate that graph-based architectures significantly improve the ability to identify hidden transactional relationships and coordinated laundering structures compared to traditional machine learning techniques. However, several limitations remain insufficiently addressed, including heterogeneous graph modeling, real-time transaction analysis, institutional privacy preservation, and scalability across distributed financial environments. Most existing approaches either rely on centralized transaction repositories or focus primarily on banking or cryptocurrency transaction datasets without effectively modeling the complex trade relationships associated with Trade-Based Money Laundering activities.

Furthermore, many existing AML frameworks process financial transactions in isolation and fail to capture the interconnected dependencies among importers, exporters, shell entities, logistics providers, and trade documents distributed across multiple jurisdictions. The absence of privacy-preserving collaborative intelligence mechanisms further limits the practical deployment of such systems in real-world financial ecosystems where direct sharing of sensitive transactional data is restricted by regulatory policies. In addition, limited focus has been given to integrating heterogeneous graph learning, federated intelligence, and real-time anomaly detection within a unified Anti-Money Laundering framework. These research gaps highlight the need for an intelligent, scalable, and privacy-preserving

architecture capable of modeling complex trade ecosystems while supporting collaborative real-time TBML detection across distributed financial institutions.

1.3 Motivation

The work, “*Federated Heterogeneous Graph Neural Networks for Real-Time Detection of Trade-Based Money Laundering*”, was chosen to address the increasing complexity of financial crimes within modern global trade ecosystems through the development of an intelligent and privacy-preserving fraud detection framework. Conventional Anti-Money Laundering (AML) systems primarily rely on rule-based monitoring and isolated transaction analysis methods that are often ineffective in identifying hidden relationships and coordinated laundering activities distributed across interconnected financial networks. Although such systems are widely deployed in banking environments, they generate high false positive rates, require extensive manual investigation, and face significant challenges in detecting sophisticated Trade-Based Money Laundering (TBML) activities in real time.

The proposed work aims to overcome these limitations by developing a Federated Heterogeneous Graph Neural Network framework capable of modeling complex transactional relationships among financial entities while preserving institutional privacy. The integration of heterogeneous graph learning, graph attention mechanisms, and federated intelligence enables collaborative fraud analysis without direct sharing of sensitive transactional data across organizations. In addition, the framework supports automated Suspicious Transaction Report (STR) generation and dashboard-based visualization for real-time fraud monitoring and compliance analysis, demonstrating the feasibility of scalable and AI-driven Anti-Money Laundering systems for modern financial environments.

1.4 Problem Statement

Existing Anti-Money Laundering (AML) systems primarily rely on rule-based monitoring and isolated transactional analysis, making them ineffective in detecting complex Trade-Based Money Laundering (TBML) activities and hidden financial relationships. In addition, privacy and regulatory constraints restrict collaborative fraud detection across financial institutions. Therefore, there is a need for a scalable and privacy-preserving framework capable of modeling interconnected financial data and accurately identifying suspicious transactional behavior in real time.

1.5 Objectives

The objectives of the project are

1. To develop a Federated Heterogeneous Graph Neural Network framework for real-time detection of Trade-Based Money Laundering activities.
2. To model complex transactional relationships among financial entities using heterogeneous graph representation and graph attention mechanisms.
3. To generate Suspicious Transaction Reports (STRs) for identified fraudulent transactions to support financial investigation and compliance processes.
4. To design an interactive dashboard for visualization of fraud predictions, transaction analysis, and real-time monitoring of suspicious activities.

1.6 Brief Methodology of the project

The proposed methodology focuses on developing a privacy-preserving and graph-based framework for real-time Trade-Based Money Laundering (TBML) detection. Initially, financial transaction data, trade records, customer information, and entity relationships are collected and preprocessed to remove inconsistencies and prepare structured datasets for analysis. The processed data is then transformed into a heterogeneous graph structure where entities such as accounts, organizations, invoices, and transactions are represented as interconnected nodes and edges.

Graph Attention-based learning mechanisms are employed to capture hidden relational dependencies and suspicious transactional patterns within the financial ecosystem. To preserve institutional privacy, federated learning is integrated into the framework, enabling collaborative model training across distributed financial environments without direct sharing of sensitive transactional data. The detected suspicious activities are further utilized for automated Suspicious Transaction Report (STR) generation and visualization through an interactive dashboard for real-time monitoring and fraud analysis.

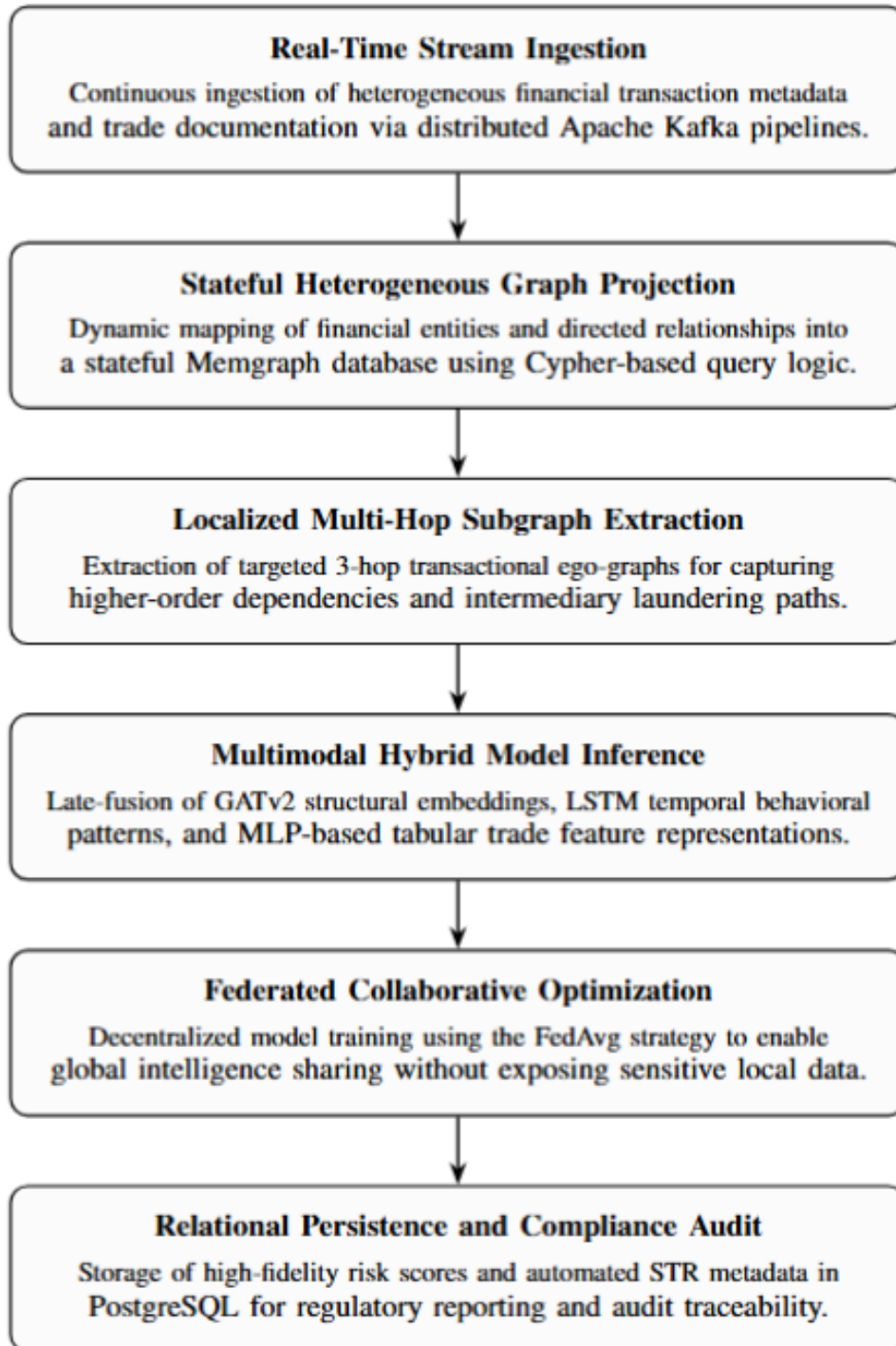


Figure 1.1: Proposed Methodology for TBML Detection

1.7 Assumptions and Constraints of the Work

Assumptions

- The financial transaction records, trade invoices, and customer-related information used for analysis are assumed to be sufficiently structured, consistent, and preprocessed for effective model training and evaluation.
- It is assumed that participating financial institutions maintain similar transactional formats and standardized trade-related attributes required for graph construction and suspicious activity analysis.
- The generated financial datasets are assumed to contain sufficient normal and suspicious transactional patterns necessary for learning hidden behavioral relationships associated with Trade-Based Money Laundering activities.
- It is assumed that heterogeneous graph representations can effectively model relationships among accounts, organizations, transactions, invoices, and trade entities for identifying suspicious financial behavior.

Constraints

- Limited access to real-world Trade-Based Money Laundering datasets due to institutional privacy regulations and confidentiality policies may affect the diversity of transactional data used for evaluation.
- Large-scale graph construction and real-time transaction analysis may increase computational complexity and memory requirements during fraud detection operations.
- The performance of the proposed framework depends significantly on the quality, completeness, and accuracy of the financial data available for model training and testing.
- The proposed framework is validated primarily under controlled academic and experimental conditions and may require additional optimization for large-scale industrial deployment.

1.8 Organization of the Report

This report is organized as follows.

- Chapter 2 discusses the theoretical concepts related to Anti-Money Laundering systems, Trade-Based Money Laundering, Graph Neural Networks, heterogeneous graph learning, and federated learning techniques used in the proposed work.
- Chapter 3 discusses the Software Requirement Specification (SRS) of the proposed system, including functional requirements, performance requirements, software and hardware requirements, and system constraints.
- Chapter 4 discusses the high-level architecture and overall system design of the proposed framework, including workflow diagrams, system components, dashboard architecture, and data flow representation.
- Chapter 5 discusses the detailed design of the proposed system, including graph construction methodology, fraud detection modules, STR generation workflow, and dashboard integration.
- Chapter 6 discusses the implementation details of the proposed framework, including dataset preprocessing, graph generation, backend integration, model development, and dashboard implementation.
- Chapter 7 discusses the testing and validation procedures performed on the proposed system, including functional testing, fraud prediction analysis, dashboard testing, and model evaluation techniques.
- Chapter 8 discusses the experimental results and performance evaluation of the proposed framework using various fraud detection metrics and comparative analysis.
- Chapter 9 discusses the conclusion of the proposed work, existing limitations, and possible future enhancements for improving the framework.

Chapter 2

Theory and Fundamentals of Project

CHAPTER 2

THEORY AND FUNDAMENTALS OF PROJECT

Developing an intelligent framework for Trade-Based Money Laundering detection requires a strong foundation in financial crime analysis, graph theory, and machine learning techniques. This chapter presents the fundamental concepts associated with Trade-Based Money Laundering, Graph Neural Networks, Graph Attention mechanisms, heterogeneous graph representation, and federated learning. The chapter further discusses the computational principles involved in relational transaction modeling, privacy-preserving learning, and real-time fraud analytics. These concepts collectively form the theoretical foundation for the proposed fraud detection framework.

2.1 Trade-Based Money Laundering

Trade-Based Money Laundering (TBML) is a sophisticated form of financial crime in which illicit funds are transferred and concealed through seemingly legitimate international trade activities. Unlike conventional laundering methods that primarily depend on direct financial transactions, TBML exploits import-export systems, trade invoices, shipping documents, and cross-border commercial operations to disguise the movement of illegal funds. Due to the increasing implementation of strict Anti-Money Laundering (AML) regulations and Know Your Customer (KYC) policies, criminal organizations have increasingly shifted toward trade-oriented laundering mechanisms that are comparatively difficult to detect using traditional monitoring systems.

The complexity of modern global trade ecosystems, combined with the involvement of multiple financial entities and transactional relationships, has made TBML detection a major challenge for financial institutions and regulatory agencies. Consequently, intelligent analytical techniques capable of identifying hidden transactional dependencies and suspicious financial behavior have become increasingly important in modern Anti-Money Laundering systems.

2.1.1 Introduction to TBML

Trade-Based Money Laundering primarily involves the manipulation of international trade transactions to illegally transfer value between entities located across different countries. Criminal organizations often misuse trade documentation, invoice values, shipment quantities, and commercial agreements to move illicit funds while making the transac-

tions appear legitimate within global financial systems. Since these laundering activities are embedded within regular import-export operations, identifying suspicious behavior becomes highly challenging for financial institutions and regulatory agencies.

In many TBML operations, colluding buyers and sellers intentionally manipulate the declared value of traded goods to transfer illegal financial assets across borders. Techniques such as over invoicing, under invoicing, phantom shipments, and multiple invoicing are commonly used to conceal the movement of illicit proceeds. Such laundering mechanisms exploit the complexity and high transactional volume of international trade systems, making traditional rule-based Anti-Money Laundering approaches comparatively ineffective in detecting hidden financial relationships.



Figure 2.1: Trade-Based Money Laundering through trade misinvoicing and open-account transactions

Source: Adapted from “The Role of Misinvoicing in the Money Laundering Cycle,” ResearchGate.

Figure ?? illustrates a typical Trade-Based Money Laundering workflow involving collusion between exporters and importers across different jurisdictions.

2.1.2 Common Techniques Used in TBML

Trade-Based Money Laundering activities are generally performed through the manipulation of trade documentation, shipment details, pricing structures, and financial transactions associated with international trade operations. Criminal organizations exploit the complexity of global trade ecosystems to transfer illicit value across borders while making the transactions appear legitimate. Several laundering techniques are commonly used within modern trade systems to disguise the movement of illegal funds and avoid detection from financial monitoring authorities.

Over Invoicing

Over invoicing is a laundering technique in which the value of exported or imported goods is intentionally declared higher than the actual market value. In this process, the buyer transfers an inflated payment amount to the seller, allowing excess funds to be moved across borders under the appearance of legitimate trade transactions. Criminal organizations use this technique to transfer illicit financial assets while avoiding suspicion from conventional banking systems.

Under Invoicing

Under invoicing involves declaring a lower value for traded goods when compared to their actual market price. This technique allows the buyer or seller to secretly transfer additional financial value outside formal banking channels. Under invoicing is commonly used to evade taxes, conceal illegal profits, and facilitate hidden movement of illicit assets across jurisdictions.

Multiple Invoicing

Multiple invoicing refers to the repeated use of the same trade invoice to justify several financial transactions for a single shipment of goods. By duplicating invoices across multiple banking channels or financial institutions, criminal entities can move large amounts of illicit funds while creating the appearance of legitimate trade payments. Detecting such duplicated transactional behavior is challenging in traditional rule-based monitoring systems.

Phantom Shipments

Phantom shipment laundering occurs when financial transactions are conducted for goods that are never actually shipped or delivered. Fraudulent trade documentation and

fabricated shipping records are used to create the illusion of legitimate trade activity while transferring illicit funds between colluding entities. Since no physical goods are exchanged, identifying phantom shipment activities requires strong transactional verification and relational analysis mechanisms.

Misrepresentation of Goods

Misrepresentation of goods involves intentionally providing false descriptions regarding the type, quantity, quality, or category of traded products within commercial documents and invoices. Criminal organizations use this technique to manipulate trade valuations and conceal suspicious financial activities under legitimate business transactions. Such inconsistencies are difficult to identify using isolated transactional analysis methods due to the large volume and diversity of international trade operations.

2.1.3 Challenges in Detecting TBML

Detecting TBML activities is highly challenging due to the complexity and interconnected nature of modern international trade ecosystems. Unlike conventional financial fraud, TBML transactions are often embedded within legitimate commercial operations, making suspicious activities difficult to distinguish from regular trade behavior. Criminal organizations exploit multiple jurisdictions, shell companies, intermediaries, and layered transactional structures to conceal the movement of illicit funds within normal trade systems.

Traditional AML systems primarily depend on rule-based monitoring approaches and isolated transactional analysis techniques. Although these systems are capable of identifying predefined suspicious patterns, they often fail to capture hidden relational dependencies among entities involved in large-scale trade networks. The increasing volume of cross-border financial transactions and evolving laundering strategies further reduce the effectiveness of conventional fraud detection mechanisms.

Another major challenge involves institutional privacy and regulatory compliance constraints associated with financial data sharing. Many organizations are unable to directly exchange sensitive transactional information due to confidentiality policies and legal restrictions. Consequently, modern TBML detection systems require intelligent, scalable, and privacy-preserving analytical frameworks capable of modeling complex financial relationships while supporting real-time fraud detection within large trade ecosystems.

2.2 Algorithms Used

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Federated Learning

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 User Interface

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ .

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 3

Analysis/Design of Pipelined Analog to Digital converter

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks."

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects
2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the experiment(Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table ?? is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. ??, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. ??

$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.

Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should be brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes

Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```

BIBLIOGRAPHY

- [1] IMTF Compliance Technologies, *When trade becomes a cover: Unpacking trade-based money laundering*, 2024. [Online]. Available: <https://imtf.com/blog/when-trade-becomes-a-cover-unpacking-trade-based-money-laundering/>
- [2] Financial Action Task Force and Egmont Group, “Trade-based money laundering: Trends and developments,” FATF, Paris, France, Tech. Rep., 2020. [Online]. Available: <https://fatf-gafi.org>
- [3] United Nations Office on Drugs and Crime, *Money laundering and Illicit financial flows overview*, 2024. [Online]. Available: <https://unodc.org>
- [4] Europol, *Criminal finances, money laundering and asset recovery*, 2024. [Online]. Available: <https://europa.eu>
- [5] J. Saenz and J. Lewer, “Quantifying illicit financial flows and trade misinvoicing in developing economies,” *Journal of Financial Crime Research*, vol. 18, no. 2, pp. 112–128, 2024.
- [6] R. Z. Z. L. W. Y. D. N. K.-Y. L. Jiani Fan Lwin Khin Shar, “Deep learning approaches for anti-money laundering on mobile transactions: Review, framework, and directions,” *IEEE Internet of Things Journal*, vol. 13, pp. 10 407–10 428, 2026, ISSN: 2327-4662. DOI: 10.1109/JIOT.2026.3653434
- [7] A. I. Rasmus Ingemann Tuffveson Jensen, “Fighting money laundering with statistics and machine learning,” *IEEE Access*, vol. 11, pp. 8889–8903, 2023, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2023.3239549
- [8] J. P. Y. G. G. Y. Jie Song Sijia Zhang, “Dimension expansion for learning money laundering activities hidden in transaction network,” *IEEE Transactions on Computational Social Systems*, vol. 13, pp. 251–262, 2026, ISSN: 2329-924X. DOI: 10.1109/TCSS.2025.3599534
- [9] P. Z. J. P. Y. G. G. Y. Jie Song Sijia Zhang, “Illicit social accounts? anti-money laundering for transactional blockchains,” *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 391–404, 2025, ISSN: 1556-6021. DOI: 10.1109/TIFS.2024.3518068

- [10] C. J. Zhong Li Xueting Yang, “Multi-view graph-based hierarchical representation learning for money laundering group detection,” *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 2035–2050, 2025, ISSN: 1556-6021. DOI: 10.1109/TIFS.2025.3529321
- [11] E. S. Abdul Haluk Batur Gezici, “Pagerank-based unsupervised deep vertex representations for anti-money laundering detection,” *IEEE Access*, vol. 13, pp. 196 935–196 950, 2025, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2025.3634197
- [12] S. A. C. P. D. P. A. M. Md. Rezaul Karim Felix Hermsen, “Scalable semi-supervised graph learning techniques for anti money laundering,” *IEEE Access*, vol. 12, pp. 50 012–50 029, 2024, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2024.3383784
- [13] B. Y. Kyungmo Koo Minyoung Park, “A suspicious financial transaction detection model using autoencoder and risk-based approach,” *IEEE Access*, vol. 12, pp. 68 926–68 939, 2024, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2024.3399824